

Environmental hazard: definition; Types, causes and solutions, biological hazards (COVID-19)

Environmental hazard

An environmental hazard is a substance, state or event which has the potential to threaten the surrounding natural environment or adversely affect people's health, including pollution and natural disasters such as storms and earthquakes.

It can include any single or combination of toxic chemical, biological, or physical agents in the environment, resulting from human activities or natural processes, that may impact the health of exposed subjects, including pollutants such as heavy metals, pesticides, biological contaminants, toxic waste, industrial and home chemicals.



Hazards can be categorized in three types:

- Chemical
- Biological
- Nuclear

This section deals with biological hazards, its causes, ways of encountering them and preventive measures that we can undertake for avoid or minimize the hazards. With this regard, we need to identify the hazard and assess the environment for the presence of hazards. This step is called environmental hazard identification.

Environmental hazard identification is the first step in risk assessment, which is the process of assessing the likelihood, or risk, of adverse effects resulting from a given hazard.

Risk

In simplest of terms, risk is the possibility of something bad happening. Risk is the uncertainty about the occurrence of a certain event such as injury, disease, death, economic loss, or damage. It is usually expressed as a mathematical statement about the likelihood of the occurrence of the event, or in other words, it is expressed in terms of mathematical probabilities.

Biological hazards

Biological hazards, also known as biohazards, refer to biological substances that pose a threat to the health of living organisms, primarily that of humans. This can include medical waste or samples of a microorganism, viruses, or toxins (from a biological source) that can affect human health.

Biological health hazards include bacteria, viruses, parasites and moulds or fungi. They can pose a threat to human health when they are inhaled, eaten or come in contact with skin. They can cause illness such as food poisoning, tetanus, respiratory infections or parasite infection.



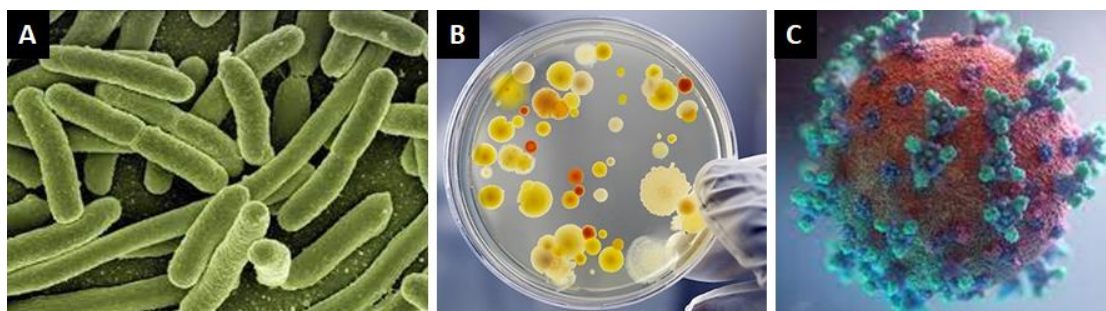
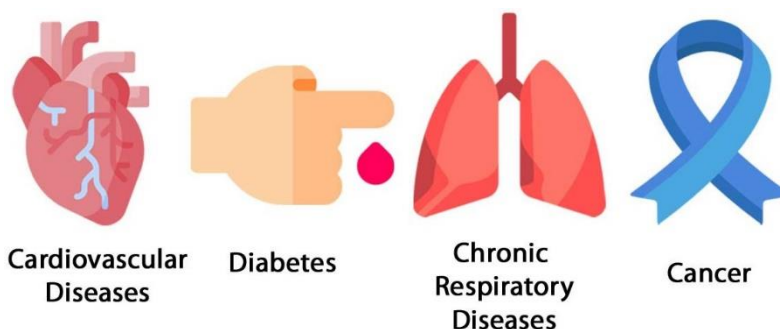


Image showing A) Bacteria, B) Mold/yeast, and C) Viruses

The main source of biological hazards is due to diseases caused by various factors. These diseases can be classified into transmissible and non-transmissible diseases.

Non-transmissible diseases (NCD)

A non-communicable disease (NCD) is a disease that is not transmissible directly from one person to another. NCDs include Parkinson's disease, autoimmune diseases, strokes, most heart diseases, most cancers, diabetes, chronic kidney disease, osteoarthritis, osteoporosis, Alzheimer's disease, cataracts, and others. NCDs may be chronic or acute. Most are non-infectious, although there are some non-communicable infectious diseases, such as parasitic diseases in which the parasite's life cycle does not include direct host-to-host transmission. NCDs are the leading cause of death globally. In 2012, they caused 68% of all deaths (38 million) up from 60% in 2000.



Transmissible/Communicable diseases

Communicable diseases, also known as infectious diseases or transmissible diseases, are illnesses that result from the infection, presence and growth of pathogenic (capable of causing disease) biologic agents in an individual human or other animal host. These diseases spread from one person to another through a variety of ways that include: contact with blood and bodily fluids; breathing in an airborne virus; or by being bitten by an insect. Some examples of the reportable communicable diseases include Hepatitis A, B & C, influenza, measles, and salmonella, tuberculosis, COVID-19, Ebola and several others.



Certain communicable diseases can spread at different rates and to varying geographical locations, resulting in either an endemic, epidemic or a pandemic disease.

Endemic disease

An endemic disease is consistently present but limited to a particular region. This makes the disease spread and rates predictable. Malaria, for example, is considered endemic in certain countries and regions.

Epidemic disease

An epidemic is the rapid spread of disease to a large number of patients among a given population within an area in a short period of time. Yellow fever, smallpox, measles, and polio are prime examples of epidemics. An epidemic disease doesn't necessarily have to be contagious.

Pandemic disease

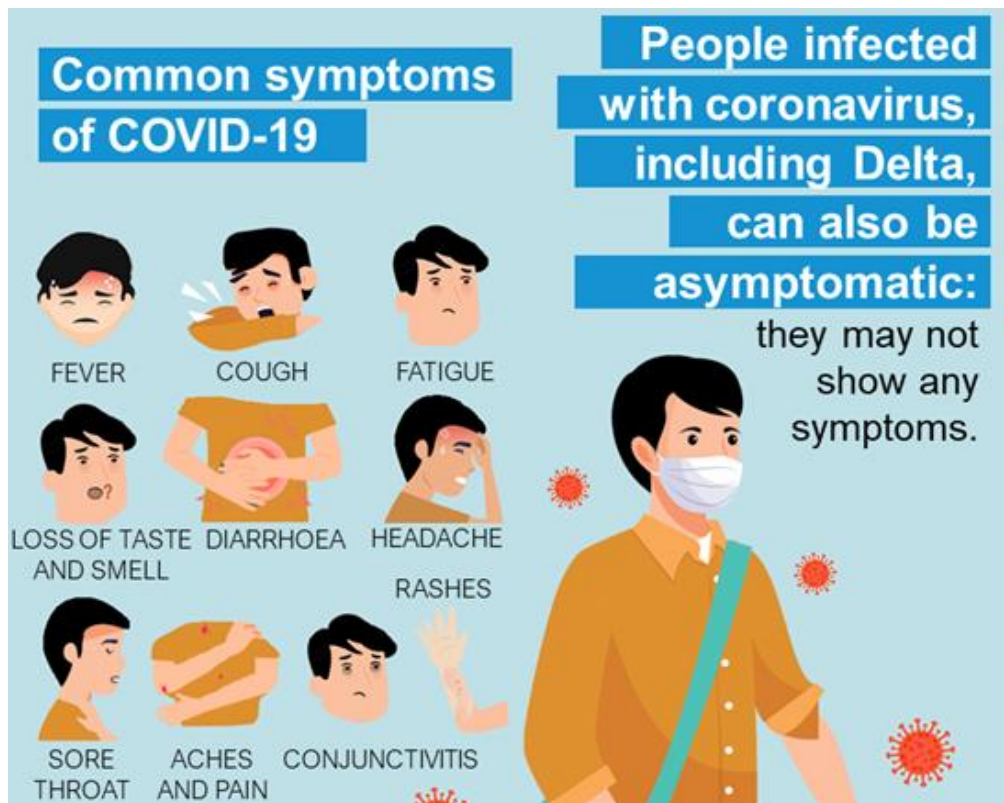
A pandemic is an epidemic of an infectious disease that has spread across a large region, for instance multiple continents or worldwide, affecting a substantial number of individuals. Recent pandemics include tuberculosis, Russian flu, Spanish flu, Asian flu, cholera, Hong Kong flu, HIV/AIDS, and COVID-19.

COVID-19

Coronavirus disease (COVID-19) is an infectious disease caused by the SARS-CoV-2 virus. The first known case was identified in Wuhan, China, in December 2019. The disease quickly spread worldwide, resulting in the COVID-19 pandemic.

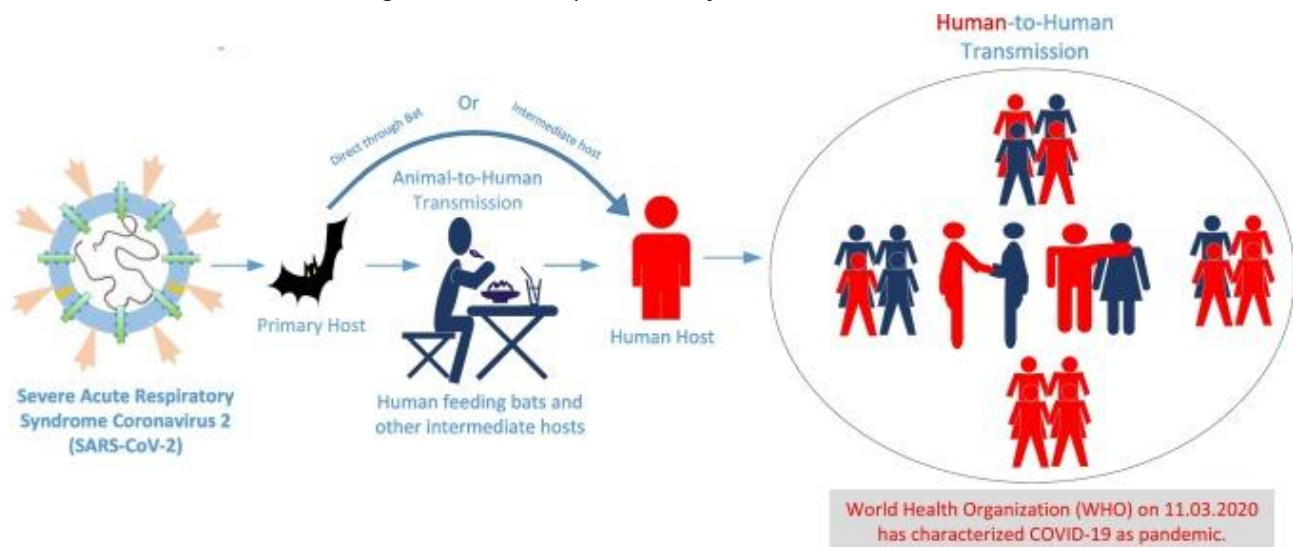
Symptoms

Symptoms of COVID-19 are variable, but often include fever, cough, headache, fatigue, breathing difficulties, loss of smell, and loss of taste. Symptoms may begin one to fourteen days after exposure to the virus. At least a third of people who are infected do not develop noticeable symptoms. Of those people who develop symptoms noticeable enough to be classed as patients, most (81%) develop mild to moderate symptoms (up to mild pneumonia), while 14% develop severe symptoms (dyspnea, hypoxia, or more than 50% lung involvement on imaging), and 5% develop critical symptoms (respiratory failure, shock, or multiorgan dysfunction). Older people are at a higher risk of developing severe symptoms. Some people continue to experience a range of effects (long COVID) for months after recovery, and damage to organs has been observed.



Transmission

COVID-19 is mainly transmitted when people breathe in air contaminated by droplets/aerosols and small airborne particles containing the virus. Infectious particles range in size from aerosols that remain suspended in the air for long periods of time to larger droplets that remain airborne briefly or fall to the ground. Infected people exhale those particles as they breathe, talk, cough, sneeze, or sing. Transmission is more likely the more physically close people are. However, infection can occur over longer distances, particularly indoors.



SARS-CoV-2 Variants

As of December 2021, there are five dominant variants of SARS-CoV-2 spreading among global populations:

- ✚ Alpha variant (B.1.1.7, formerly called the UK variant)
- ✚ Beta variant (B.1.351, formerly called the South Africa variant)
- ✚ Gamma variant (P.1, formerly called the Brazil variant)
- ✚ Delta variant (B.1.617.2, formerly called the India variant)
- ✚ Omicron variant (B.1.1.529)

Treatment

Most people who become sick with COVID-19 will only have mild illness and can get better at home. Symptoms might last a few days. People who have the virus might feel better in about a week. Several treatment options are available to people with coronavirus (COVID-19) who are at the highest risk of becoming seriously ill. The treatments available are:

- ✚ Nirmatrelvir And Ritonavir (Paxlovid)
- ✚ Sotrovimab (Xevudy)
- ✚ Remdesivir (Veklury)
- ✚ Molnupiravir (Lagevrio)

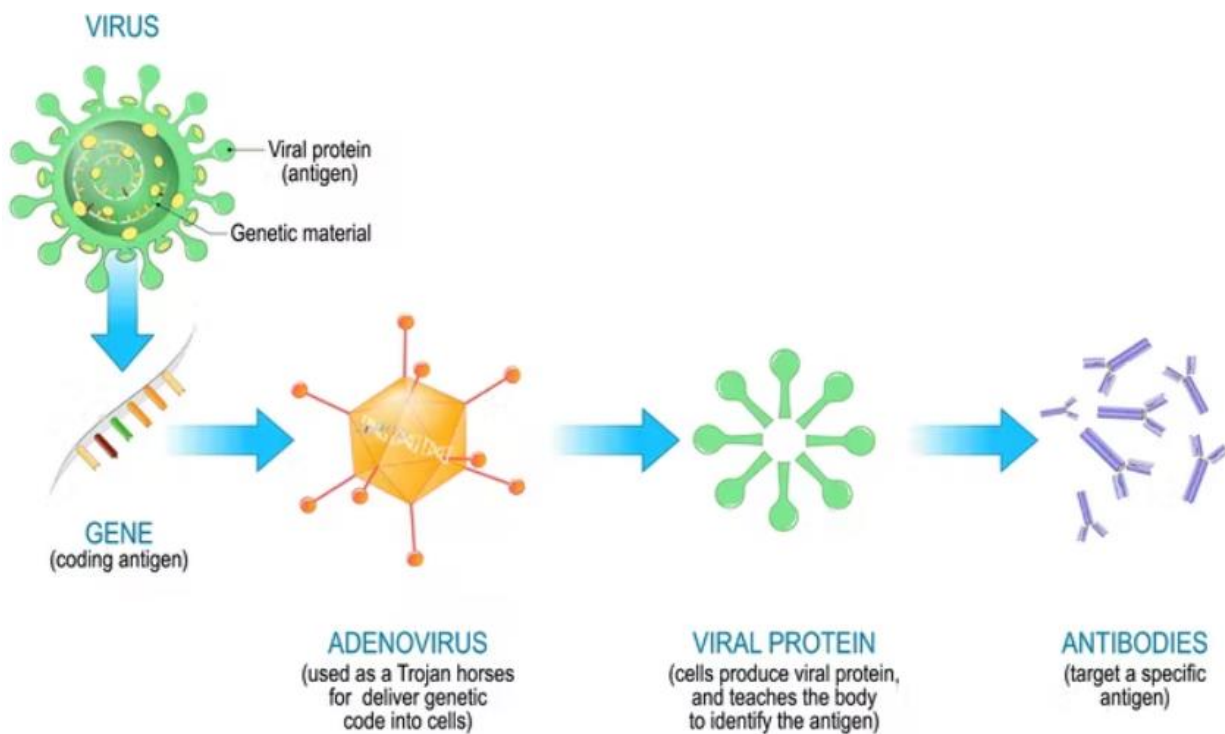
Vaccines for COVID-19

A COVID-19 vaccine is a vaccine intended to provide acquired immunity against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), the virus that causes coronavirus disease 2019 (COVID-19). Mass vaccination programmes have been established by WHO, and nine vaccines have been approved for emergency or full use by at least one stringent regulatory authority recognized by the World Health Organization (WHO): Pfizer-BioNTech, Oxford-AstraZeneca, Sinopharm BIBP, Moderna, Janssen, CoronaVac, Covaxin, Novavax, and Convidecia. Each of these vaccines causes the immune system to create antibodies for fight COVID-19 using a harmless version of a spike-like structure on the surface of the COVID-19 virus. These vaccines act by different mechanisms, a few of which are explained below. The different types of vaccines include:

- ✚ Viral vector vaccines
- ✚ mRNA vaccines
- ✚ Whole virus vaccines
- ✚ Protein sub-unit vaccines

Viral vector vaccine

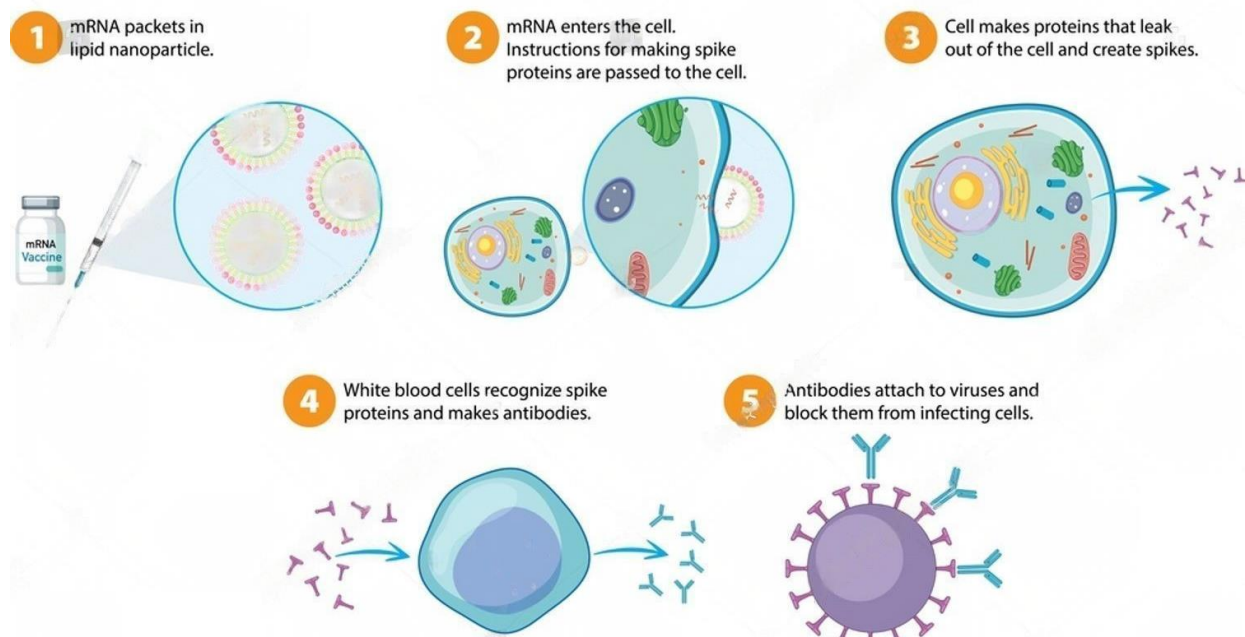
In this type of vaccine, genetic material from the COVID-19 virus is placed in a modified version of a different virus (viral vector). When the viral vector gets into your cells, it delivers genetic material from the COVID-19 virus that gives your cells instructions to make copies of the S protein. Once your cells display the S proteins on their surfaces, your immune system responds by creating antibodies and defensive white blood cells. If you later become infected with the COVID-19 virus, the antibodies will fight the virus. The Janssen/Johnson & Johnson COVID-19 vaccine is a vector vaccine. AstraZeneca and Covaxin vaccines work on this principle.



Schematic representation for the mechanism of action of the viral vector vaccine of COVID-19

mRNA vaccines

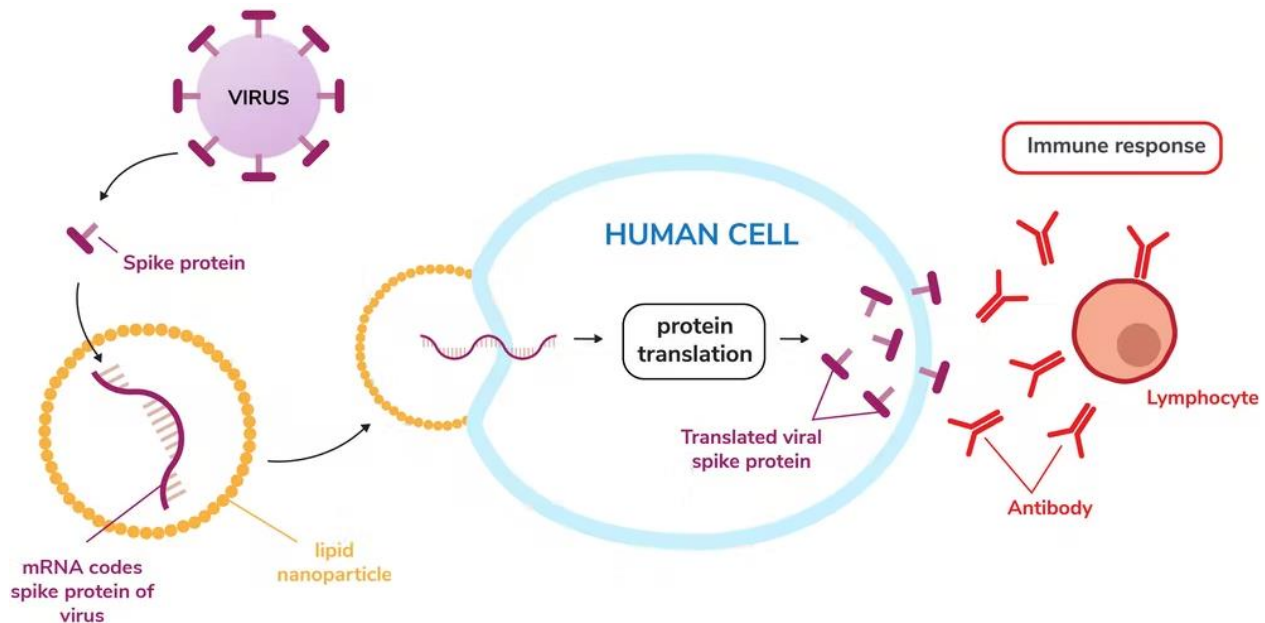
This type of vaccine uses genetically engineered mRNA to give your cells instructions for how to make the S protein found on the surface of the COVID-19 virus. After vaccination, your muscle cells begin making the S protein pieces and displaying them on cell surfaces. This causes your body to create antibodies. If you later become infected with the COVID-19 virus, these antibodies will fight the virus. Both the Pfizer-BioNTech and the Moderna COVID-19 vaccines use mRNA.



Schematic representation for the mechanism of action of mRNA vaccine of COVID-19

Whole virus vaccines

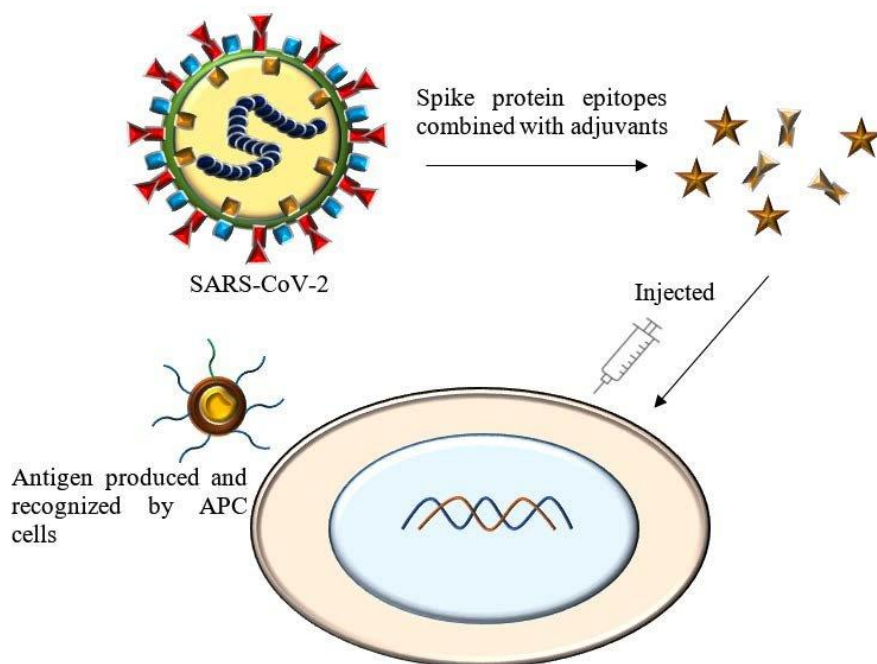
Whole virus vaccines use a weakened (attenuated) or deactivated form of the pathogen that causes a disease to trigger protective immunity to it. the advantages of an inactivated whole virus vaccine include the fact its technology is well established, it is suitable for people with compromised immune systems, and it's relatively simple to manufacture.



Schematic representation for the mechanism of action of the whole virus vaccine of COVID-19

Protein subunit vaccine

Subunit vaccines include only the parts of a virus that best stimulate your immune system. This type of COVID-19 vaccine contains harmless S proteins. Once your immune system recognizes the S proteins, it creates antibodies and defensive white blood cells. If you later become infected with the COVID-19 virus, the antibodies will fight the virus. Novavax is working on a protein subunit COVID-19 vaccine. Covishield is one such vaccine prepared according to this method.



Schematic representation for the mechanism of action of the protein subunit vaccine of COVID-